Mapping Lake Baringo's Rising Shoreline, 2002–2026, and Where Not to Build

Thelma Cherop | Department of Mechatronic Engineering, School of Engineering, Dedan Kimathi University of Technology
Brian Lochuny | Institute of Geomatics, GIS and Remote Sensing (IGGReS), Dedan Kimathi University of Technology

Scan for the maps, data and method
github.com/brianlochuny-cloud/lake_baringo_project

Background

Lake Baringo, in Kenya's Rift Valley, has expanded sharply since the 2010s, affecting lodges, camps and shoreline communities. Planners need to know how far the lake has grown, whether growth is continuing, and which land is at risk. This study measures the lake from satellite images for six years (2002–2026) and maps projected footprints for 2027 and 2032 to show where infrastructure should be avoided.

Methods

- Landsat Collection 2 Level-2 scenes (USGS): Landsat 7 (2002, 2007, 2012), Landsat 8 (2017, 2022), Landsat 9 (2026); dry season, mainly February
- Striping in the 2007 and 2012 Landsat 7 scenes filled using a second scene
- Water mapped with MNDWI = (Green − SWIR1) / (Green + SWIR1); water where MNDWI > 0
- Area measured in ArcGIS Pro (WGS 84 / UTM 37N), keeping water connected to the 2002 lake
- 2027 and 2032 footprints from a smoothed SRTM 1 arc-second DEM, with the elevation level matched to each scenario's area
- Exposure counted from OpenStreetMap buildings, roads and facilities (minimum counts)

99% overall accuracy, kappa 0.98
2026 water map, 100 random points checked against the satellite image (same image as the map, so true accuracy may be slightly lower)

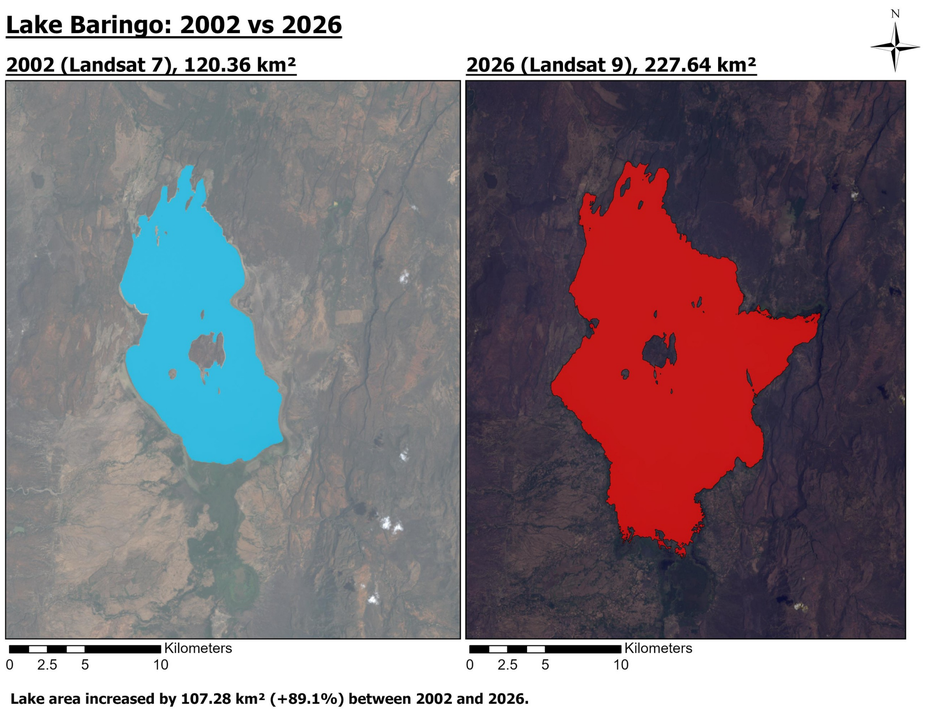


Figure 1. Lake Baringo in 2002 and 2026: +107.28 km² (+89.1%).

Results: six years of change

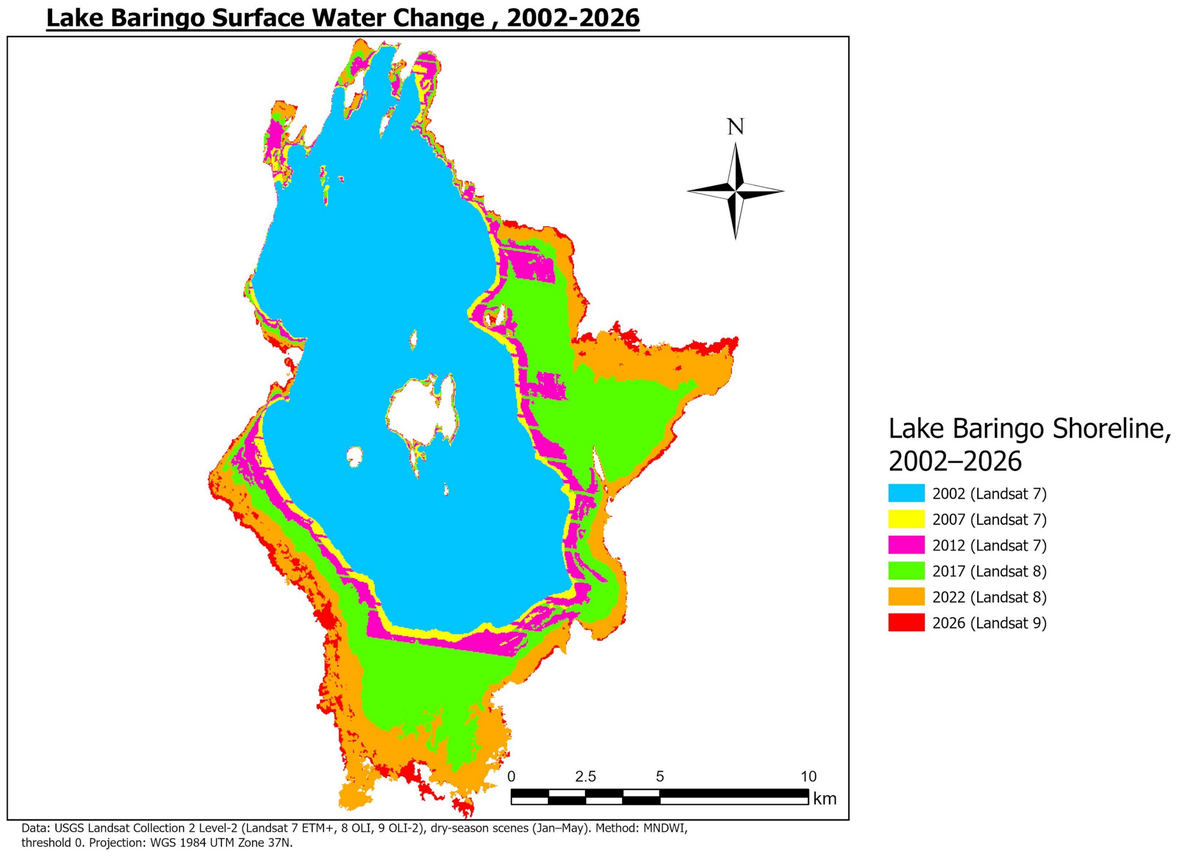


Figure 2. Shoreline of Lake Baringo, 2002–2026 (observed).

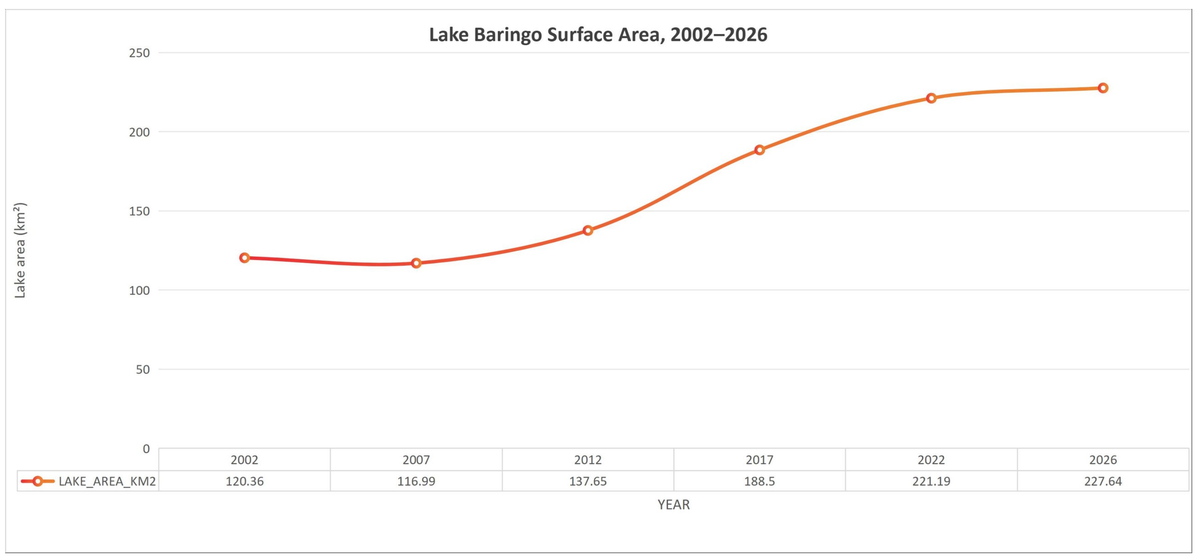
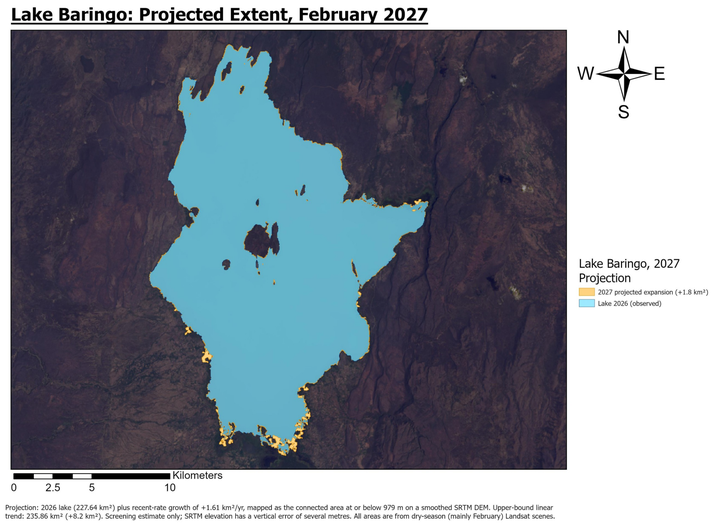
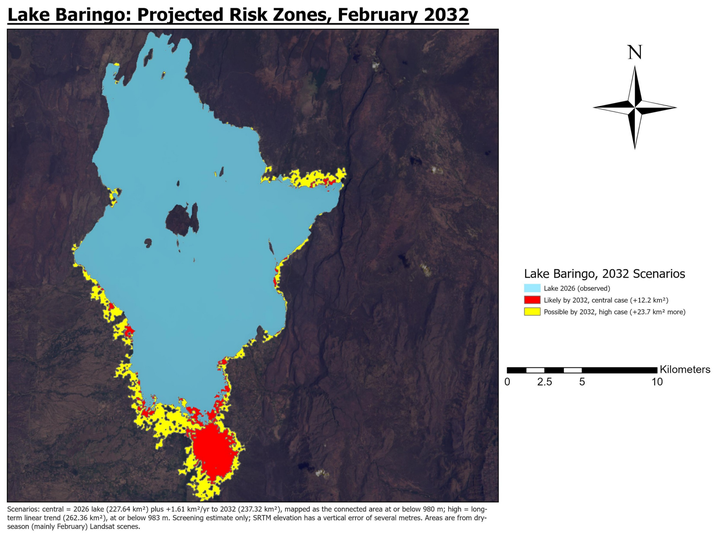


Figure 3. Lake area, 120.36 km² (2002) to 227.64 km² (2026).

Period	Change (km²)	Rate (km²/yr)
2002–2007	−3.37	−0.67
2007–2012	+20.66	+4.13
2012–2017	+50.85	+10.17
2017–2022	+32.69	+6.54
2022–2026	+6.45	+1.61

Projected expansion and risk zones (screening estimate)





Figures 4–5. Projected extent for February 2027 (left; +1.8 km²) and risk zones for February 2032 (right; central +12.2 km², high a further +23.7 km²).

Mapped exposure (minimum counts)

Scenario	Bldgs	Roads (km)
2027	17	2.2
2032 central	27	3.9
2032 high	115	26.8

Facilities: Korosin Campsite (2027 band); one more toilet in the outer ring. Power lines not assessed.

Conclusions

- Lake Baringo grew 89% in 24 years, 120.36 to 227.64 km²
- Growth is slowing: about 10 km²/yr in 2012–2017, about 1.6 km²/yr in 2022–2026
- 2027 estimate: about 229 km²; 2032: about 228 (low), 237 (central), 262 km² (high)
- Planners should avoid the 2032 central zone and treat the high zone with caution

Limitations and data

- SRTM (2000) has several metres of vertical error; 1 m of level changes the area by about 10 km²
- 2027 and 2032 are projections from six data points, not rainfall-based forecasts
- OpenStreetMap is incomplete in rural areas; counts are minimums
- Data: USGS Landsat C2 L2 and SRTM; OpenStreetMap contributors. Analysis in ArcGIS Pro